

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D Deterministic SHE-Boltzmann MOSFET Simulation

Complete report, Revision 2 — coupled, impact-ionization-in-BTE state

Reproduction study (code commit 891e7c9; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

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Abstract

This report derives and numerically reproduces the deterministic spherical-harmonic (SHE) Boltzmann device solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997). Revision 2 supersedes the earlier one-way/post-processed description: impact ionization is now *inside* the assembled electron collision operator (primary depletion + secondary redistribution), the self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole-continuity loop is *demonstrated and observably converging*, global number- and terminal-current-conservation diagnostics are in place, and a physics-split preconditioner removes the previous solver bottleneck (~ 10 min $\rightarrow \sim 15$ s on the 40×34 problem). Established results: core transport (n , $|v|$, T_e) is validated and spatially robust; the self-consistent feedback materially suppresses avalanche generation and drain current while leaving T_e comparatively stable; conservation holds across one-way, refined, and coupled runs. Open items are stated precisely: the impact-ionization magnitude is *not spatially converged* — a 2×2 grid $\times$ coupling study drives $G_{ii,\max}$ multiplicatively from 1.35×10^{27} to $5.24 \times 10^{26} \text{ cm}^{-3}\text{s}^{-1}$ (the finest-tested coupled value, not a continuum estimate), while a one-way energy refinement to $\Delta H = 6.25 \text{ meV}$ moves it only -2.4% (negligible beside the spatial error); the T_e/G_{ii} peak offset is unresolved; and the device has now been recalibrated to $V_{th} = 0.614 \text{ V}$ for the pending Liang overlays. The robust reproductions are the transport moments and the ($\sim 37\%$) self-consistent G_{ii} -feedback fraction — the latter now *confirmed* on the recalibrated $V_{th} = 0.614 \text{ V}$ device (-36%), where the higher overdrive raises the absolute (still grid-unconverged) generation to 1.34×10^{27} . Every model element is classified as *paper-specified*, *reconstruction*, *surrogate*, or *not-yet-converged*.

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1 Overview and audit provenance

Liang *et al.* solve, self-consistently, three coupled equations on a 2-D MOSFET cross-section: Poisson, the electron BTE (first-order SHE), and hole continuity with an impact-ionization pair-generation source. This reproduction has passed through four external audit rounds and three code patches (Appendix B); the present revision documents the resulting coupled state.

Honesty note. Model substitutions carried throughout and classified in Table 1: the doping profile is a *reconstruction*; the band is a 1-parameter Kane *surrogate* (paper: 2-band Fiegna/Brunetti); the impact-ionization rate is a Keldysh *surrogate*; the momentum time τ_1 is a single calibrated surrogate (not a summed Table-I model); and the device carries an un-recalibrated $V_{th} = 1.64$ V.

Part I

Method derivation and validated foundation

2 First-order SHE equation

With $\mathbf{E} = -\nabla_{\mathbf{r}}\phi$, expand $f = f_0(\mathbf{r}, \varepsilon) + \hat{\mathbf{\Omega}} \cdot \mathbf{f}_1(\mathbf{r}, \varepsilon)$ (all $\ell \geq 2$ discarded). Projection gives the current relation and the conservative augmented-space balance

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z \mathcal{Q}[f_0] + Z S_{\text{gen}}, \quad (1)$$

$\mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0)$, $D = \frac{1}{3} v^2 \tau_1$. In total energy $H = \varepsilon - q\phi$ this becomes pure spatial diffusion at fixed H , the collision operator coupling $H \leftrightarrow H \pm E_{op}$ locally in space, over the global mesh $-q\phi_{max} \leq H \leq \varepsilon_{max} - q\phi_{min}$ with mask $0 \leq H + q\phi \leq \varepsilon_{max}$; $\varepsilon = 0$ reflecting, ε_{max} absorbing. The DOS-weighted acoustic operator $\mathcal{Q}_{ac} = Z^{-1} \partial_{\varepsilon} [ZA(\partial_{\varepsilon} + 1/k_B T) f_0]$ (Scharfetter–Gummel) makes the collision term irreducible; it is number-conserving (defect 7.6×10^{-17}) with the Maxwellian as its exact null.

3 Reconstructed device, taxonomy, foundation

The $0.35 \mu\text{m}$ LDD NMOS uses error-function junctions ($x_j = 0.170 \mu\text{m}$ metallurgical). Foundation checks (all at $T = 300$ K): equilibrium Poisson (built-in $+0.586 / -0.44$ V, quadratic

Newton); nonparabolic $N_c = 2.857 \times 10^{19}$ (0.1% of the independent value; parabolic limit 0.09% of closed form); bulk scattering engine detailed-balance 2×10^{-5} , zero-field $T_e \rightarrow 309.7$ K, monotone heating. The DD solve at $V_{gs} = V_{ds} = 3$ V gives the operating-point potential driving the SHE.

Table 1: Modelling taxonomy (Rev. 2 adds a *not-yet-converged* class).

Element	Class	Basis / status
Method eqs., SHE reduction, moments	paper	Liang <i>et al.</i>
Table-I params, spherical band	paper	paper Sec. II
$L_{\text{eff}}, t_{ox}, x_j, N_D^\pm$	paper	paper text
1-param Kane $\gamma(\varepsilon)$	surrogate	paper: 2-band Fiegna/Brunetti
LDD doping shape, mesh	reconstruction	ref. [7] unavailable
Channel $N_A, \Phi_{gate}, \tau_1, c_{op}, 1/\tau_{ii}$	surrogate	calibrated; disclosed
$G_{ii, \text{max}}$ magnitude	not-yet-converged	spatially unconverged; 5.24×10^{26}
T_e/G_{ii} peak offset	not-yet-converged	(coupled 60×50), still dropping
		unresolved at present mesh

Part II

The 2-D SHE core, impact ionization, and self-consistency

4 Assembled operator, impact ionization, and moments

The solver discretizes the $(\mathbf{r}, H)$ equation by box integration; contributions (spatial diffusion $\times \Delta H$, optical in/out $\times A_{ij} \Delta H$, acoustic SG flux $\times A_{ij}$) share units and form a positive-diagonal system. Contacts inject a Maxwellian $f_0 = n_{\text{contact}} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$.

Impact ionization in the electron BTE. An event removes a primary electron at rate $1/\tau_{ii}(\varepsilon)$ (Keldysh surrogate, $\varepsilon_{th} = 1.1$ eV) and deposits two secondaries at $\varepsilon' = (\varepsilon - \varepsilon_{th})/2$, linearly distributed onto the neighbouring H nodes, with net electron gain +1 per event. This is *assembled into the electron collision operator* (primary depletion + secondary redistribution), so f_0 is directly shaped by avalanche; $G_{ii} = \int Z f_0 / \tau_{ii} d\varepsilon$ is a moment of that II-affected distribution. The absorbing ε_{max} cutoff acts on spatial, acoustic, and optical fluxes. Moments n , $\langle \mathbf{v} \rangle$, T_e , G_{ii} , and the terminal currents are reconstructed from the *same finite-volume face fluxes* as the operator.

5 Numerics: physics-split preconditioner

The large-grid system is solved by a multiplicative physics-split preconditioner. Because DOFs are numbered with H fastest, same-spatial-cell couplings (acoustic + optical + II redistribution + spatial diagonal) form contiguous local energy blocks — the **E block** — factored cheaply; the diagonal plus cross-cell couplings form the fixed- H real-space **S block**. One application does $E^{-1} \rightarrow S^{-1} \rightarrow E^{-1}$ with a fresh residual between stages; the solved matrix is never shifted (shifts touch only preconditioner factors). On the 328 544-DOF 40×34 problem this solves to 10^{-10} in ~ 15 s (raw scaled residual 1.22×10^{-11}), where the previous global ILU did not finish its setup in 120 s. The legacy path is retained as `method="legacy_ilu"` for regression.

6 Numerical audit of the one-way baseline

Table 2 is the extracted audit of the locked one-way reference (commit `f016e3c`); it is preserved as a regression benchmark.

Table 2: Clean 40×34 numerical audit (impact ionization in the collision operator).

Quantity	Value	Location / note
$n_{\max}$	$1.0516 \times 10^{20} \text{ cm}^{-3}$	source n^+ (0.082, 0.043)
$T_{e,\max}$	3232.9 K	drain (0.595, 0.005)
$G_{ii,\max}$	$1.3513 \times 10^{27} \text{ cm}^{-3} \text{ s}^{-1}$	drain (0.595, 0.009)
$ v _{\max}$	$9.732 \times 10^6 \text{ cm/s}$	drain surface (0.554, 0)
$T_e - G_{ii}$ peak separation	5 nm (~ 1 cell)	co-located within mesh
SHE currents I_{src}/I_{drn}	$-1.7044/ + 1.7071 \times 10^{-4} \text{ A}/\mu\text{m}$	mismatch 0.16%
global continuity residual	-5.70×10^{-6} (relative)	—
orphan states	0	no artificial sinks
f_0 pre-clip min	-3.3×10^{-11}	46/328544 tiny negatives

7 Grid refinement (40×34 vs 60×50 , one-way)

Refining the spatial mesh at fixed $\Delta H = 12.5 \text{ meV}$:

Quantity	40×34	60×50	converged?
$n_{\max}$	1.0516×10^{20}	1.0473×10^{20}	yes (-0.4%)
$ v _{\max}$	9.73×10^6	10.0×10^6	yes ($+2.7\%$)
$T_{e,\max}$ [K]	3232.9	3423.0	moderate ($+5.9\%$)
$G_{ii,\max}$	1.3513×10^{27}	8.26×10^{26}	no (-40%)
T_e/G_{ii} separation	1 cell	2 cells	grid-scale (unresolved)
current mismatch / residual	0.16% / 5.7×10^{-6}	0.07% / 7.6×10^{-6}	conserved

Transport (n , $|v|$, T_e) is spatially robust; G_{ii} is *not* grid-converged (it depends on the high-energy tail); and the peak separation tracks the mesh, so it is unresolved, not a reproduced physical offset.

8 Self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole loop

A damped outer iteration solves the SHE (with impact ionization) on the current potential, drives hole continuity with G_{ii} as pair generation (holding n_{SHE}), updates Poisson from the SHE density, damps, and repeats; n is recomputed from the SHE after each potential update (not constitutively from DD). Over 12 iterations (Fig. 2) the maximum potential update falls monotonically $17.7 \rightarrow 4.0 \text{ mV}$, and the physical observables plateau: $G_{ii,\max}$ is suppressed 37% ($1.35 \rightarrow 0.85 \times 10^{27}$) and the drain current 29% ($1.71 \rightarrow 1.21 \times 10^{-4} \text{ A}/\mu\text{m}$), while $T_{e,\max}$ stays $\sim 3240 \text{ K}$. This is the physically expected negative feedback: the SHE electrons and the G_{ii} -generated holes reshape the potential, lowering the peak drain field and throttling avalanche. Final conservation is clean (relative number balance 2.4×10^{-5} , currents balance 0.14%).

Honesty note. The loop is *demonstrated and observably converging*, not converged to the prescribed tolerance: the plateaued observables are meaningful, but $\max |\Delta\phi| \approx 4 \text{ mV}$ remains above the chosen 2 mV stopping criterion. Tighter convergence needs more iterations or reduced damping.

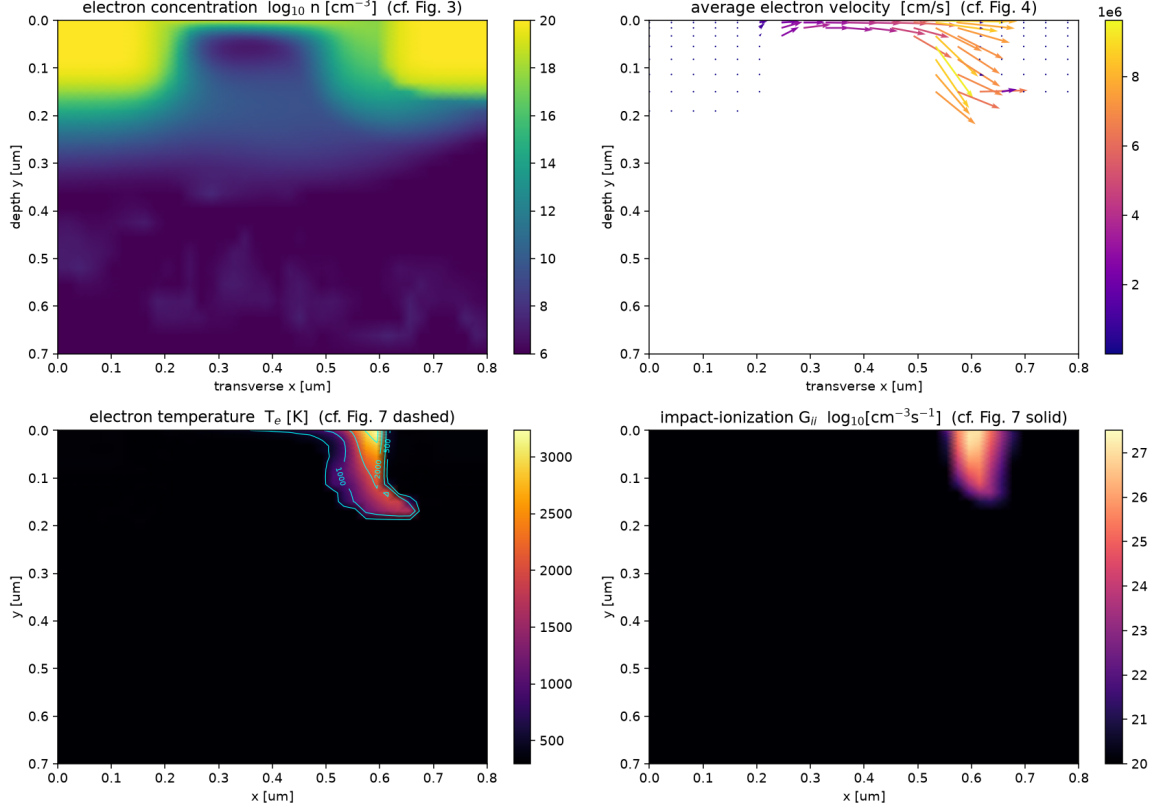


Figure 1: One-way SHE moments (40×34). n (Fig. 3), velocity field saturating at 9.73×10^6 cm/s (Fig. 4; masked where $n < 10^{13}$ cm $^{-3}$), T_e (Fig. 7 dashed, drain-localized), G_{ii} (Fig. 7 solid, drain-localized).

9 G_{ii} convergence matrix and device recalibration

Combining one-way and coupled runs at both grids isolates how the impact-ionization magnitude responds to spatial refinement and to self-consistency (Table 3). The spatial ratios ($8.26/13.5 = 0.612$ one-way, $5.24/8.50 = 0.616$ coupled) and the coupling ratios ($8.50/13.5 = 0.630$ at 40×34 , $5.24/8.26 = 0.634$ at 60×50) agree to $\sim 1\%$ (ratio-of-ratios 1.0085): the two effects are *multiplicatively separable to that precision* over the tested meshes, and $1.35 \times 10^{27} \times 0.612 \times 0.630 = 5.20 \times 10^{26}$ predicts the observed 5.24×10^{26} . The coupled result is *still* spatially unconverged (coupled $40 \times 34 \rightarrow 60 \times 50$: -38% , no plateau), and, like the 40×34 loop, the coupled 60×50 outer iteration ended above the 2 mV max $|\Delta\phi|$ criterion (observably converging, not formally converged).

Table 3: $G_{ii,\max}$ [cm $^{-3}$ s $^{-1}$] over grid $\times$ coupling.

	one-way	self-consistent
40×34	1.35×10^{27}	8.50×10^{26}
60×50	8.26×10^{26}	5.24×10^{26}

A *one-way* energy refinement to $\Delta H = 6.25$ meV ($E_{op} = 8$) moves $G_{ii,\max}$ only -2.4% ($T_e -0.19\%$) — *negligible beside* the -38 – 40% spatial error — so the dominant numerical uncertainty is resolving the drain-side high-field/high-energy hotspot in real space rather than the energy axis. (The energy test was performed one-way, not on the coupled configuration in which the spatial sensitivity was measured, so we state energy insensitivity, not a proof of purely-spatial error.) 5.24×10^{26} is thus the *finest-tested coupled value*, not an estimate of the

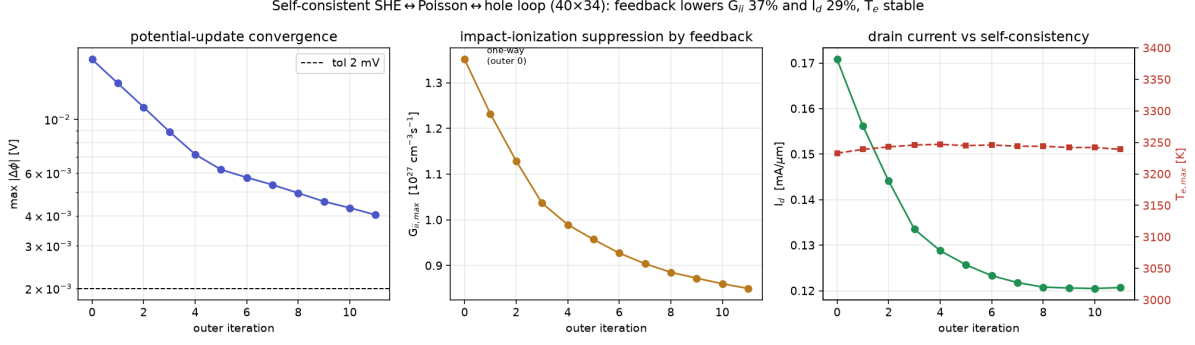


Figure 2: Self-consistent loop (40×34). Potential update converges $17.7 \rightarrow 4.0$ mV; feedback suppresses G_{ii} 37% and I_d 29% (both plateauing) while T_e stays stable.

continuum limit: with a -38% change under the last spatial refinement there is no numerical basis to quote a converged G_{ii} . The robust results are the transport moments and the ($\sim 37\%$, grid-independent) feedback fraction.

Device recalibration. A rigid gate offset $\Phi_{gate} = -0.74$ V ($dV_{th}/d\Phi_{gate} \approx +1$) recalibrates the extracted threshold to $V_{th} = 0.614$ V (was 1.64 V); the $V_g = 3$ V overdrive becomes 2.39 V. The coupled 60×50 operating point on this device converges the observables ($n_{max} = 1.01 \times 10^{20}$, $T_{e,max} = 3598$ K, $I_d = 0.24$ mA/μm, currents balance 0.02%, final $\max |\Delta\phi| = 7.5$ mV) and *preserves the G_{ii} -feedback fraction* (-36% vs. -37% on the old device; current suppression -55% at the higher overdrive). The higher overdrive raises the absolute $G_{ii,max}$ to 1.34×10^{27} ($\sim 2.5\times$ the old-device coupled value, from more inversion/heating) — device-dependent and still grid-unconverged. Digitized Liang overlays on this recalibrated operating point are the outstanding quantitative work.

Part III

Assessment

10 Conclusions, classified

- **Core SHE transport** shows a clean grid-sensitivity hierarchy: n strongly robust (0.4%), $|v|$ robust (2.7%), T_e moderately grid-sensitive (5.9%; robust but not converged), G_{ii} unconverged (40%).
- **Impact ionization:** now part of the assembled electron collision operator, with explicit primary depletion and secondary redistribution (G_{ii} is a moment of the II-affected f_0).
- **Self-consistency:** SHE ↔ Poisson ↔ hole feedback *demonstrated and observably converging* (not to the 2 mV criterion); it materially suppresses avalanche generation and drain current while leaving T_e comparatively stable. The G_{ii} -suppression fraction is *confirmed* on the recalibrated $V_{th} = 0.614$ V device (-36% ; current suppression is stronger, -55% , at the higher 2.39-V overdrive).
- **Conservation:** validated across one-way, refined, and coupled calculations (number balance 10^{-5} – 10^{-7} ; terminal currents balance $< 0.2\%$).
- **Numerics:** physics-split preconditioning removes the previous practical solver bottleneck.
- G_{ii} : the earlier 1.37×10^{27} agreement with Liang was *demonstrably accidental* — spatial refinement and self-consistency each push it downward (multiplicatively separable to $\sim 1\%$).

The coupled 60×50 run gives $5.24 \times 10^{26} \text{ cm}^{-3} \text{ s}^{-1}$, the *finest-tested coupled value* (not a continuum estimate: still -38% below coupled 40×34 , no plateau); a one-way energy refinement moves it only -2.4% . *No converged magnitude can be quoted* (Table 3).

- **T_e/G_{ii} offset:** unresolved at present mesh resolution; *no longer claimed reproduced*.
- **Device:** recalibrated to $V_{th} = 0.614 \text{ V}$ ($\Phi_{gate} = -0.74 \text{ V}$), replacing the earlier over-high 1.64 V .
- **Remaining quantitative gaps:** digitized Liang target-figure overlays on the recalibrated coupled solution, and (lower priority) a spatially-converged G_{ii} via drain-local refinement.

A Numerical and model specifications

$T = 300 \text{ K}$, Boltzmann, $m^* = 0.32m_0$, $g_v = 6$, $\alpha = 0.5 \text{ eV}^{-1}$, $v_s = 9.04 \times 10^3 \text{ m/s}$; erfc doping per earlier revisions; SHE grid $\Delta H = 12.5 \text{ meV}$ ($E_{op}/\Delta H = 4$ integer), $\varepsilon = 0$ reflecting / ε_{max} absorbing. Optical $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$, $\tau_1 = 10^{-14} \text{ s}/(\sqrt{\gamma}\gamma')$, $\tau_{ac} = \tau_1$; impact ionization $1/\tau_{ii} = P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2 \Theta$, $P = 2 \times 10^{13} \text{ 1/s}$, $\varepsilon_{th} = 1.1 \text{ eV}$, equal-split secondaries. Solve: physics-split $E \rightarrow S \rightarrow E$ preconditioner, LGMRES tol 10^{-10} (audit) / 10^{-8} (coupled iterations). Coupled loop: ϕ -damping 0.25 , ≤ 12 outer iterations, 2 mV stop.

B Provenance and audit history

Git commit `891e7c9`; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Key commits: `f016e3c` (locked one-way audit), `7aad998` (60×50 refinement), `65461ad` (physics-split solver), `047da93` (coupled loop). Four audit rounds fixed: over-claims, missing specs, junction geometry, N_c convergence, the reducible-collision zero-field- T_e bug (acoustic operator), the global H -grid, the contact-Maxwellian normalization (report typo; code correct), boundary/velocity/current extraction, and finally delivered the real impact-ionization operator, the coupled driver, and the physics-split solver.

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